Techinical details on the implementation

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1 Heap asymptotics

We heapify the container by levels starting from the bottom one. Let n be the Heap size. Then $(n-1)/2$ is the index of the first node having left child. As we assume that all the substrees starting from the level $i+1$ are valid min heaps, we have that the tree with the root at index j at the level i may only have the root out of order, which is fixed by sinking it down. The number of executions of the sinking down is given by:

$$\sum_{i=0}^{h-1}(2^i \cdot (h-i))$$

where h is the height of the heap, 2^i is the number of nodes at the level i and $h-i$ is the number of level below the i-th one.

Lemma 1.1. *For any $q \in (-1,1)$:*

$$\sum_{k=0}^{\infty} kq^k = \frac{q}{(1-q)^2}$$

Proof. Suppose $\sum_{k=1}^{\infty} kq^k$ converges for $|q|<1$. Let $S_n = \sum_{k=1}^{n} kq^k$:

$$\begin{aligned}
S_n - qS_n = \sum_{k=1}^{n} kq^k - \sum_{k=1}^{n} kq^{k+1} &= \sum_{k=1}^{n} kq^k - \sum_{k=2}^{n+1}(k-1)q^k \\
&= q + \sum_{k=2}^{n} kq^k - \sum_{k=2}^{n}(k-1)q^k - nq^{n+1} \\
&= q + \sum_{k=2}^{n}(k-k+1)q^k - nq^{n+1} \\
&= \sum_{k=1}^{n} q^k - nq^{n+1}
\end{aligned}$$

Since we assume the sequence S_n to converge, $nq^{n+1} \to 0$ as a sequence must converge to zero to be summable.

$$\begin{aligned}
(1-q)\lim_{n\to\infty} S_n = \lim_{n\to\infty}(S_n - qS_n) &= \lim_{n\to\infty}\left(\sum_{k=1}^{n} q^k - nq^{n+1}\right) \\
&= \frac{q}{1-q} - \lim_{n\to\infty} nq^{n+1} \\
&= \frac{q}{1-q}
\end{aligned}$$

Therefore:

$$\sum_{k=1}^{\infty} kq^k = \frac{q}{(1-q)^2}$$

Consequently:

$$\sum_{k=0}^{\infty} kq^k = 0 \cdot q^0 + \sum_{k=1}^{\infty} kq^k = \frac{q}{(1-q)^2}$$

To prove convergence of S_n we can use the ratio test:

$$\left|\frac{(k+1)q^{k+1}}{kq^k}\right| = |q|\frac{k+1}{k}$$

The sequence $(k+1)/k \to 1$. Since $|q| < 1$ we can pick θ in such a way that $|q| < \theta < 1$. Hence we have that there exists $n_0 \in \mathbb{N}$, s.t. $|(k+1)/k - 1| = (k+1)/k - 1 < 1/\theta - 1$ for all $n > n_0$

$$|q|\frac{k+1}{k} \leq \frac{|q|}{\theta} < 1 \quad \forall n \geq n_0$$

By substituting $k = h - i$ we get that the sum is equal to:

$$\sum_{i=0}^{h-1}(2^i \cdot (h-i)) = \sum_{k=1}^{h}(2^{h-k} \cdot k) = 2^h \sum_{k=1}^{h} 2^{-k} \cdot k < 2^h \sum_{k=0}^{\infty}(2^{-k} \cdot k) = 2^h \cdot \frac{1/2}{(1-1/2)^2}$$
$$= 2 \cdot 2^h \leq 2n$$

where we have used that the number of nodes in a complete binary heap is $2^{h+1} - 1$ hence $n \geq 2^h$ as the levels $1, \ldots, h-1$ are full and there is at least one node at the level h.

2 Linear regression

Assume the following construction:

$$y_i = \beta^T x_i + \varepsilon_i, \quad \varepsilon_1, ..., \varepsilon_n \overset{iid.}{\sim} \mathcal{N}(0, \sigma^2)$$

where $x_i \in \mathbb{R}^d$ and $\beta \in \mathbb{R}^d$ are fixed

2.1 Preliminaries

Lemma 2.1. *Let $X : \Omega \to \mathbb{R}^d$ be a random variable and $A \in \mathbb{R}^{d\times d}, b \in \mathbb{R}^d$. Assume that the second moments exist, i.e. $\mathbb{E}[X_i^2] < \infty$ and the covariance matrix is given by Σ. Then the following properties hold:*

Property 2.1.1 $\mathbb{E}[AX + b] = A \cdot \mathbb{E}[X] + b$

Property 2.1.2 $\mathrm{Cov}(AX) = A \cdot \Sigma \cdot A^T$

Remark 2.1. Using the fact that the second moment exists and by Jensen's inequality, we have that $\mathbb{E}[X_i] < \infty$ and $\mathbb{E}[X_i^2] < \infty$ for all $i = 1, ..., d$. Therefore the covariance matrix is well-defined, since $(\mathbb{E}[|X_i X_j|])^2 = (\mathbb{E}[|X_i||X_j|])^2 \leq \mathbb{E}[|X_i|^2]\,\mathbb{E}[|X_j|^2] < \infty$ by Cauchy-Schwarz inequality.

Proof.

1. Trivial

2. Suppose that the expected value of X is the zero vector. By Property 2.1.1 we also have that $\mathbb{E}[AX] = A\,\mathbb{E}[X] = 0$. Then we have that:

$$\begin{aligned}
\operatorname{Cov}((AX)_i, (AX)_j) &= \operatorname{Cov}\left(\sum_{k=1}^{d} A_{ik}X_k, \sum_{l=1}^{d} A_{jl}X_l\right) = \mathbb{E}\left(\left(\sum_{k=1}^{d} A_{ik}X_k\right)\left(\sum_{l=1}^{d} A_{jl}X_l\right)\right) \\
&= \mathbb{E}\left[\sum_{k=1}^{d}\sum_{l=1}^{d} A_{ik}X_kX_lA^T_{lj}\right] = \sum_{k=1}^{d}\sum_{l=1}^{d} A_{ik}\,\mathbb{E}[X_kX_l]A^T_{lj} \\
&= \sum_{k=1}^{d} A_{ik}\left(\sum_{l=1}^{d} \Sigma_{kl}A^T_{lj}\right) = \sum_{k=1}^{d} A_{ik}(\Sigma A^T)_{kj} = (A\Sigma A^T)_{ij}
\end{aligned}$$

If the expected value of X is not the zero vector, we can define a new random variable $Y = X - \mathbb{E}[X]$, which has the expected value of zero. Then we have that:

$$\operatorname{Cov}(Y_i, Y_j) = \mathbb{E}[Y_iY_j] = \mathbb{E}[(X_i - \mathbb{E}[X]_i)(X_j - \mathbb{E}[X]_j)] = \operatorname{Cov}(X_i, X_j)$$

Therefore we have that:

$$\begin{aligned}
\operatorname{Cov}((AX)_i, (AX)_j) &= \mathbb{E}[(AX - \mathbb{E}[AX])_i \cdot (AX - \mathbb{E}[AX])_j] \\
&\overset{*}{=} \mathbb{E}[(A(X - \mathbb{E}[X]))_i \cdot (A(X - \mathbb{E}[X]))_j] \\
&= \mathbb{E}[(AY)_i \cdot (AY)_j] = \operatorname{Cov}((AY)_i, (AY)_j) \\
&= (A\operatorname{Cov}(Y)A^T)_{ij} = (A\operatorname{Cov}(X)A^T)_{ij}
\end{aligned}$$

where $*$ follows from Property 2.1.1.

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Lemma 2.2 (Multivariate normal distribution)**.** *We say that $X : \Omega \to \mathbb{R}^d$ is multivariately normally distributed if and only if there exists a matrix $A \in \mathbb{R}^{d\times d}$ and a vector $b \in \mathbb{R}^d$ such that $X = AZ + b$, where $Z_1, ..., Z_d \overset{iid.}{\sim} \mathcal{N}(0, 1)$.*

We have that multivariate normal distribution is closed under linear transformations, i.e. if $X \sim \mathcal{N}(\mu, \Sigma)$ and $Y = AX + b$, then $Y \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$. (For the expectation and covariance refer to Property 2.1.1 and Property 2.1.2).

Lemma 2.3. *For the multivariate normal distribution, we have that the components are independent if and only if they are uncorrelated, i.e. $\operatorname{Cov}(X_i, X_j) = 0$ for all $i \neq j$.*

Lemma 2.4. *Let $X \in \mathbb{R}^{n\times d}$ be a matrix of full rank, where $n > d$. Define $H = X(X^TX)^{-1}X^T \in \mathbb{R}^{n\times n}$ and $E = \operatorname{diag}\{\underbrace{1, ..., 1}_{n-d}, \underbrace{0, ..., 0}_{d}\}$, then:*

Property 2.4.1 $I - H$ *is symmetric and idempotent.*

Property 2.4.2 $\operatorname{rank}(I - H) = n - d$

Property 2.4.3 There exists a matrix Q such that $I - H = Q^TEQ$

Property 2.4.4 $\ker(I - H) = \operatorname{im}(X)$

Proof.

1. This follows by the direct computation:

$$(I-H)^T = I - H^T = I - (X(X^TX)^{-1}X^T)^T = I - (X^T)^T\left((X^TX)^T\right)^{-1}X^T = I - H$$

$$\begin{aligned}(I-H)(I-H) &= (I - X(X^TX)^{-1}X^T)(I - X(X^TX)^{-1}X^T)\\ &= I - 2X(X^TX)^{-1}X^T + X(X^TX)^{-1}X^TX(X^TX)^{-1}X^T\\ &= I - 2X(X^TX)^{-1}X^T + X(X^TX)^{-1}X^T\\ &= I - X(X^TX)^{-1}X^T = I - H\end{aligned}$$

2. Since $I-H$ is idempotent we have that its trace is equal to its rank. Therefore we have that:

$$\text{trace}(H) = \text{trace}(X(X^TX)^{-1}X^T) \stackrel{*}{=} \text{trace}((X^TX)^{-1}X^TX) = \text{trace}(I_d) = d$$

where $*$ follows from the cyclic property of the trace. Therefore we have that:

$$\text{rank}(I-H) = \text{trace}(I-H) = \text{trace}(I) - \text{trace}(H) = n - d$$

3. Since the matrix $I-H$ is idempotent we have that it has an eigenvalue 1 with multiplicity equal to $\text{rank}(I-H) = n-d$ and eigenvalue 0 with multiplicity d. Its minimal polynomial is contained in $\{x, x-1, x(x-1)\}$ implying that $I-H$ is diagonalizable since in all the cases it does not have repeated factors. Therefore there exists a matrix $Q \in O(\mathbb{R}^n)$, s.t.

$$Q^T(I-H)Q = E$$

4. By the rank-nullity theorem we have $\dim\ker(I-H) = n - \text{rank}(I-H) = n-(n-d) = d$. Using the assumption we have that $\dim \text{im}(X) = d$. Therefore it suffices to show that $\text{im}\, X \subset \ker(I-H)$:

$$\begin{aligned}(I-H)Xy &= (I - X(X^TX)^{-1}X^T)Xy\\ &= Xy - X(X^TX)^{-1}X^TXy = Xy - Xy = 0 \quad \forall y \in \mathbb{R}^d\end{aligned}$$

This shows that $\text{im}\, X \subset \ker(I-H)$, and since they have the same dimension we have that $\text{im}\, X = \ker(I-H)$.

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2.2 t-statitistic

In this section we assume that $\hat{\beta}$ is the ordinary least squares estimator of β. We establish two facts:

- $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ since the components of ε are independent and identically distributed as $\mathcal{N}(0,\sigma^2)$, thus we can obtain $\varepsilon = \sigma I \cdot (\varepsilon/\sigma)$, where $\varepsilon_1/\sigma, ..., \varepsilon_n/\sigma \stackrel{iid.}{\sim} \mathcal{N}(0,1)$.
- $y = X\beta + \varepsilon \sim \mathcal{N}(X\beta, \sigma^2 I)$ as a consequence of Lemma 2.2 and the fact that y is a linear transformation of ε.

Step 1. Compute the distribution of the residual sum of squares:

$$\hat{\varepsilon} = y - X\hat{\beta} = y - X(X^TX)^{-1}X^Ty = (I-H)y$$

By Property 2.4.4 we have that $\mathrm{im}(X) = \ker(I-H)$, and since $X\beta \in \mathrm{im}(X)$ we have that:

$$(I-H)X\beta = 0$$

thus:

$$(I-H)y = (I-H)(X\beta + \varepsilon) = (I-H)\varepsilon$$

with the notation from Lemma 2.4. Using the result from Property 2.4.3 we have that:

$$I - H = Q^TEQ$$

for some orthogonal matrix Q.

Writing out the expression for the residual sum of squares, we have that:

$$\mathrm{RSS} = \hat{\varepsilon}^T\hat{\varepsilon} = \varepsilon^T(I-H)^T(I-H)\varepsilon = \varepsilon^T(I-H)\varepsilon$$

Define $U = Q\varepsilon$. Then $U \sim \mathcal{N}(0, \sigma^2 I)$ by Lemma 2.2. Then we have that:

- $\mathbb{E}[Q\varepsilon] = Q\,\mathbb{E}[\varepsilon] = 0$ by Property 2.1.1
- $\mathrm{Cov}(Q\varepsilon) = Q\,\mathrm{Cov}(\varepsilon)Q^T = Q\sigma^2IQ^T = \sigma^2QQ^T = \sigma^2 I$ by Property 2.1.2

Then by Lemma 2.3 we have that the components of U are independent, since they are uncorrelated.

Thus we have that U/σ follows a joint standard normal distribution. Then we have that:

$$\mathrm{RSS} = U^TEU \tag{1}$$

$$\frac{\mathrm{RSS}}{\sigma^2} = U^T/\sigma \cdot E \cdot U/\sigma = \sum_{i=1}^{n-d}(U_i/\sigma)^2 \sim \chi^2_{n-d}$$

Step 2. Compute the distribution of $\hat{\beta}_i - \beta_i$:

Observe that the $\hat{\beta}$ follows multivariate normal distribution since $\hat{\beta} = Ay$ where $A = (X^TX)^{-1}X^T$, and $y \sim \mathcal{N}(X\beta, \sigma^2 I)$ (see Lemma 2.2). Therefore $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2AA^T)$, where we have used that $\hat{\beta}$ is an unbiased estimator of β and that

$$\mathrm{Cov}(\hat{\beta}) = \mathrm{Cov}(Ay) = A\,\mathrm{Cov}(y)A^T = \sigma^2AA^T$$

by Property 2.1.2.

$$AA^T = (X^TX)^{-1}X^TX\left((X^TX)^{-1}\right)^T = \left((X^TX)^{-1}\right)^T = \left((X^TX)^T\right)^{-1} = (X^TX)^{-1}$$

As was derived above, we have that $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X^TX)^{-1})$. Therefore we have that:

$$\hat{\beta}_i - \beta_i \sim \mathcal{N}(0, \sigma^2(X^TX)^{-1}_{ii})$$

Thus:

$$\frac{\hat{\beta}_i - \beta_i}{\sigma\sqrt{(X^TX)^{-1}_{ii}}} \sim \mathcal{N}(0,1)$$

Step 3. $\hat{\varepsilon}$ and $\hat{\beta} - \beta$ are independent:

We write $\hat{\varepsilon} = (I - H)\varepsilon$ and $\hat{\beta} - \beta = (X^T X)^{-1} X^T (X\beta + \varepsilon) - \beta = (X^T X)^{-1} X^T \varepsilon$. Then we have that:

$$\begin{pmatrix} \hat{\varepsilon} \\ \hat{\beta} - \beta \end{pmatrix} = \begin{pmatrix} I - H \\ (X^T X)^{-1} X^T \end{pmatrix} \varepsilon$$

Thus by Lemma 2.2 we have that the joint distribution of $\hat{\varepsilon}$ and $\hat{\beta} - \beta$ is multivariate normal. Therefore by Lemma 2.3 it suffices to show that they are uncorrelated, i.e. $\mathrm{Cov}(\hat{\varepsilon}, \hat{\beta} - \beta) = 0$. We have that:

$$\begin{aligned} \mathrm{Cov}(\hat{\varepsilon}, \hat{\beta} - \beta) &\overset{*}{=} \mathbb{E}[\hat{\varepsilon}(\hat{\beta} - \beta)^T] = \mathbb{E}[(I - H)\varepsilon\varepsilon^T X (X^T X)^{-1}] \\ &\overset{**}{=} (I - H)\,\mathbb{E}[\varepsilon\varepsilon^T] X (X^T X)^{-1} \\ &= \sigma^2 (I - H) X (X^T X)^{-1} \end{aligned}$$

where $*$ follows from the fact that both $\hat{\varepsilon}$ and $\hat{\beta} - \beta$ have expected value of zero as the linear transformation of ε which has expected value of zero. $**$ follows from Property 2.1.1.

By Lemma Property 2.4.4 we have that $\mathrm{im}(X) = \ker(I - H)$, thus $(I - H)X = 0$, hence:

$$\mathrm{Cov}(\hat{\varepsilon}, \hat{\beta} - \beta) = \sigma^2 (I - H) X (X^T X)^{-1} = 0$$

Step 4. Conclude that the t-statistic follows a t-distribution with $n - d$ degrees of freedom:

$$\frac{\hat{\beta}_i - \beta_i}{\sqrt{\frac{\mathrm{RSS}}{n-d} (X^T X)^{-1}_{ii}}} = \frac{\frac{\hat{\beta}_i - \beta_i}{\sigma\sqrt{(X^T X)^{-1}_{ii}}}}{\frac{1}{\sigma}\sqrt{\frac{\mathrm{RSS}}{n-d}}} = \frac{\frac{\hat{\beta}_i - \beta_i}{\sigma\sqrt{(X^T X)^{-1}_{ii}}}}{\sqrt{\frac{\mathrm{RSS}}{\sigma^2 (n-d)}}} = \frac{Z}{\sqrt{\frac{T}{n-d}}}$$

where by the previous steps we have that $Z = \frac{\hat{\beta}_i - \beta_i}{\sigma\sqrt{(X^T X)^{-1}_{ii}}} \sim \mathcal{N}(0, 1)$ and $T = \frac{\mathrm{RSS}}{\sigma^2} \sim \chi^2_{n-d}$. T depends only on the residuals and Z depends only on $\hat{\beta}_i$, which are independent by the previous step. Therefore we have that the t-statistic follows a t-distribution with $n - d$ degrees of freedom.

Theorem 2.1. *The unbiased estimator of the variance of the error term is given by:*

$$\hat{\sigma}^2 = \frac{1}{n - d} \sum_{i=1}^{n} \hat{\varepsilon}_i^2$$

where d is the number of features and n is the number of samples.

Proof.

$$\mathrm{RSS} = \hat{\varepsilon}^T \hat{\varepsilon} \overset{(1)}{=} \sum_{i=1}^{n-d} U_i^2$$

Each U_i is distributed as $\mathcal{N}(0, \sigma^2)$, therefore we have that:

$$\mathbb{E}[U_i^2] = \mathrm{Var}(U_i) = \sigma^2$$

Thus we have that:

$$\mathbb{E}[\text{RSS}] = \mathbb{E}\left[\sum_{i=1}^{n-d} U_i^2\right] = \sum_{i=1}^{n-d} \mathbb{E}[U_i^2] = (n-d)\sigma^2$$

Dividing both sides by $n-d$ yields:

$$\mathbb{E}\left[\frac{\text{RSS}}{n-d}\right] = \sigma^2$$

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Remark 2.2. This provides an intuition why we use this estimator for the variance of the error term.

Theorem 2.2 (Gauss-Markov theorem)**.** *The ordinary least squares estimator $\hat{\beta}$ is the best linear unbiased estimator (BLUE) of β, i.e. for each $C \in \mathbb{R}^{d\times n}$ such that $\mathbb{E}[Cy] = \beta$, we have that:*

$$\text{Cov}(\hat{\beta}) \preceq \text{Cov}(Cy)$$

Proof. We need to show that the matrix:

$$\text{Cov}(Cy) - \text{Cov}(\hat{\beta})$$

is positive semidefinite.

Define $D = C - (X^TX)^{-1}X^T$. Then we have that:

$$\mathbb{E}[Cy] = \mathbb{E}[((X^TX)^{-1}X^T + D)y] = \mathbb{E}[(X^TX)^{-1}X^Ty] + \mathbb{E}[Dy] = \beta$$

Thus we have that $\mathbb{E}[Dy] = 0$. Therefore we have that:

$$\mathbb{E}[Dy] = \mathbb{E}[DX\beta + D\varepsilon] = DX\beta = 0$$

Since β is unknown we must have that $DX\beta = 0$ for all β, which implies that $DX = 0$. This also implies $(DX)^T = X^TD^T = 0$. Then we have that:

$$\begin{aligned}
\text{Cov}(Cy) &= \sigma^2 CC^T \\
&= \sigma^2((X^TX)^{-1}X^T + D)((X^TX)^{-1}X^T + D)^T \\
&= \sigma^2((X^TX)^{-1}X^TX(X^TX)^{-1} + DX(X^TX)^{-1} + (X^TX)^{-1}X^TD^T + DD^T) \\
&= \sigma^2((X^TX)^{-1} + DD^T)
\end{aligned}$$

We know that DD^T is positive semidefinite, so for any $v \in \mathbb{R}^d$ we have that:

$$v^TCC^Tv = v^T(X^TX)^{-1}v + v^TDD^Tv \geq v^T(X^TX)^{-1}v$$

thus:

$$v^T(CC^T - (X^TX)^{-1})v \geq 0$$

implying that $CC^T - (X^TX)^{-1}$ is positive semidefinite. Therefore we have that:

$$\text{Cov}(Cy) - \text{Cov}(\hat{\beta}) = \sigma^2(CC^T - (X^TX)^{-1}) \succeq 0$$

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